Summary in Graph

Exam Summary (GO Classes GATE CS | C-Programming | Weekly Quiz 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	10 (5 + 5)
Total Marks:	15 (5 + 10)
Exam Duration:	30 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained

Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10

* Average Stats based on first-attempt stats of Top 10% (81) out of 809 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Numerical Type Award: 1 Penalty: 0 Programming in C

Consider the number 1001 in $2's$ Complement number system _____

Your Answer: Correct Answer: -7 Not Attempted Time taken: 00min 00sec Discuss

Q #2 Numerical Type | Award: 1 | Penalty: 0 | Programming in C

Minimum Number of bits to present $+14$ in binary in $2's$ complement system?

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Consider a $2's$ complement system with 16-bit short integer size. What is the smallest value that can be stored in a signed short?

A. -2^{15}
B. $-2^{15} - 1$
C. -2^{16}
D. $-2^{16} - 1$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What will be the output of the following program?

(Assume $2's$ complement system for signed numbers)

```c
int main() {
    int i = -1;
    int x = (unsigned char)i;
    printf("%d", x);
}
```

A. -1
B. 255
C. A huge number
D. 11111111

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Multiple Select Type | Award: 1 | Penalty: 0 | Programming in C

In C programming, constant integers are considered to be signed integers by default. One way to represent them as an unsigned constant is by appending U as a suffix. For example, -1 is signed, whereas $-1\mathrm{U}$ is unsigned.

Which of the following condition(s) is/are TRUE?

A. $-3 > -4\mathrm{U}$
B. $-1\mathrm{U} > -2$
C. $-1\mathrm{U} > 0\mathrm{U}$
D. $-1 > 1$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 01sec | Discuss

Q #6 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

What will be the output on the execution of the following code segment?

```
main()
{
    unsigned num1=-1;
    signed num2=1;
    if(num1 < num2)
        printf("less");
    else if(num1>num2)
        printf("greater");
    else if(num1==num2)
        printf("equal");
}
```

A. greater
B. less
C. equal
D. error

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Consider k bit binary pattern.

- $T_{\max}$ and $T_{\min}$ are maximum and minimum signed numbers we can represent using k bits.
- $U_{\max}$ and $U_{\min}$ are maximum and minimum unsigned numbers we can represent using k bits.

Which of the following(s) is/are true for $k = 16$?

Here $|.|$ represents absolute value of a number i.e., $|r| = -r$ if $r < 0$ otherwise $|r| = r$.

A. $|T_{\min}| = T_{\max} + 1$
B. $U_{\max} = 2 * T_{\max} + 1$
C. $U_{\max} = |T_{\min}| + T_{\max} + 1$
D. $U_{\min} = |T_{\min}|$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Consider Figure A and Figure B which represent 4 bit signed and unsigned numbers respectively in $2's$ complement system.

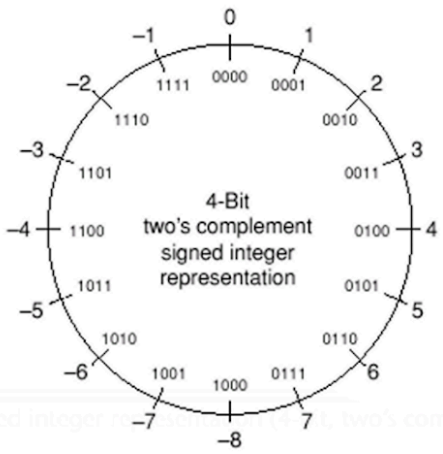

Figure A

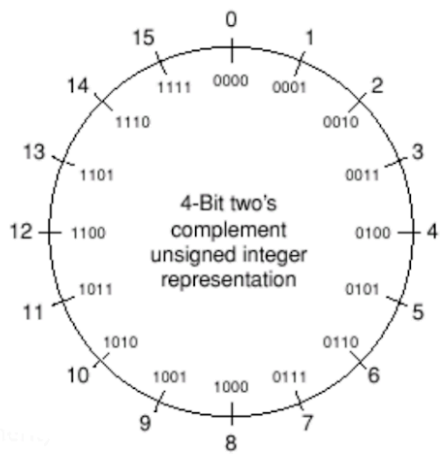

Figure B

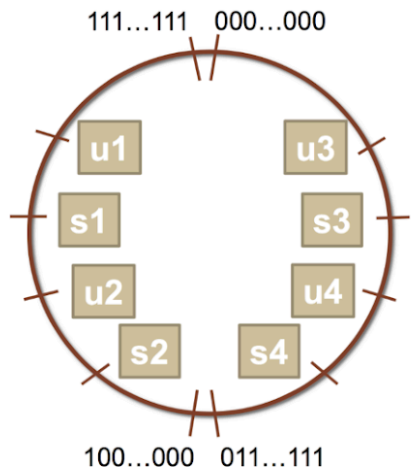

Figure C

Assume that a few variables are defined below and initialized in such a way that places them in the spot shown in Figure C.

```
int s1, s2, s3;
unsigned int u1, u2, u3;
```

Which of the following(s) is/are true?

A. $s3 > u3$
B. $s1 > s3$
C. $u1 > u3$
D. $s1 > u3$

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Consider two program fragments given below. UINT_MAX is the maximum unsigned number in the system.

Program 1

```
main()
{
 unsigned int a = UINT_MAX;
 signed char c = -1;

 if (c == a)
  printf("GO Classes");
}
```

Program 2

```
main()
{
 unsigned int b = UINT_MAX;
 signed char d = -10;

 if (d == b - 10)
  printf("GATE Overflow");
}
```

Which of the following(s) is/are true? All the prints are without double commas.

A. Program 1 prints "GO Classes"
B. Program 2 prints "GATE Overflow"
C. Program 1 does not print "GO Classes"
D. Program 2 does not print "GATE Overflow"

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

What is the output of the following code? Assume that int is 32 bits, short is 16 bits, and the representation is two's complement.

```
signed short ix = -2;
printf( "%u", ix );
```

A. -2
B. $2^{32} - 2$
C. $2^{32} - 1$
D. $2^{16} - 1$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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